

Supplementary information

Multi-view coding-style transitions for leakage-safe paired detection of LLM-paraphrased code

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All tables derive from saved evidence. No model fitting or new formal experiment was performed during this revision. F1 is positive-class F1; equal-weight means across experimental cells are not class-macro F1.

S1 Controlled contrast intervals

Contrast	Definition	Clean (pp)	95% interval	Held-out (pp)	95% interval
C1	A1 minus A0	+1.683	[1.466, 1.932]	+1.153	[1.068, 1.259]
C2	A2 minus A0	+4.592	[4.258, 4.832]	+4.976	[4.745, 5.100]
C3	A4 minus A1	+4.092	[3.923, 4.300]	+4.659	[4.499, 4.798]
C4	A5 minus A4	-0.019	[-0.079, 0.052]	+0.007	[0.000, 0.013]
C5	A5 minus A0	+5.756	[5.360, 6.003]	+5.819	[5.645, 5.938]

C1: signed differences with original features; C2: endpoint expansion; C3: endpoint expansion with signed differences; C4: relative block; C5: full representation versus original endpoints. Intervals retain folds inside three seed clusters. A positive interval for the extremely small unseen C4 is not a claim of practical importance.

S2 Source registry

Gate	Protocol	Manifest SHA-256
gate_a	all-llm-strict-origin-v2	593b2044cab542f120190d4f50bccbdc c06ff9c48cde115608f705448dd2fa95
gate_b	leave-one-llm-strict-origin-v1	10cdfea1964e78aae624fba017bd99d2 19899d07e22ed9c10e4c7ef55397c0db
gate_c	all-llm-strict-origin-attack-v1	6aaa2d77823b517a1b3ac4649cb2799 e03cf04d44045307b649526f45534383 b
gate_d	leave-one-language-strict-origin-v1	a04cb43cbf527500bd1d4cf30c833f7d 126ca25f81352b05cf7ac46b40bf4373

Full file digests are supplied in the revision provenance snapshot and frozen registry. SHA-256 values identify files, not evidence of methodological sufficiency by themselves.

S3 Per-generator and per-language results

Held-out generator	LPcode original	Full MSTF	Difference (pp)	95% interval (pp)
GPT-3.5	0.8978	0.9819	+8.412	[7.894, 8.768]
Gemini-Pro	0.8837	0.9592	+7.550	[7.196, 7.840]
WizardCoder-33B	0.9003	0.9687	+6.836	[6.463, 7.241]
DeepSeek-Coder-33B	0.8842	0.9587	+7.456	[7.236, 7.839]

Held-out language	LPcode original: mean $\pm$ SD	MSTF: mean $\pm$ SD	Difference (pp)
C	0.8922 $\pm$ 0.0046	0.9656 $\pm$ 0.0010	+7.339
C++	0.8976 $\pm$ 0.0034	0.9723 $\pm$ 0.0009	+7.475
Java	0.8799 $\pm$ 0.0076	0.9584 $\pm$ 0.0021	+7.852
Python	0.9055 $\pm$ 0.0017	0.9645 $\pm$ 0.0014	+5.897

S4 Negative-pair sensitivity

Negatives	A0 F1	A1 F1	Difference (pp)	95% interval (pp)
current	0.9149	0.9317	+1.683	[1.466, 1.932]
random	0.9119	0.9300	+1.805	[1.647, 1.896]
hard	0.8463	0.8824	+3.605	[3.431, 3.808]

These are A1–A0 comparisons with original ten features, not full MSTF. Current, random and hard refer to the registered negative construction only. Positives and isolation constraints remain fixed.

S5 Transformation uncertainty

Condition	Method	Drop (pp)	95% seed-cluster CI (pp)
combined	lpcode original	6.311	[5.822, 6.849]
combined	mstf	1.935	[1.776, 2.039]
comment injection	lpcode original	0.034	[-0.058, 0.125]
comment injection	mstf	0.119	[0.092, 0.142]
comment removal	lpcode original	0.890	[0.833, 0.979]
comment removal	mstf	0.187	[0.168, 0.212]
format normalization	lpcode original	-0.001	[-0.004, 0.003]
format normalization	mstf	0.004	[-0.001, 0.014]
identifier rename	lpcode original	5.124	[4.754, 5.543]
identifier rename	mstf	1.732	[1.642, 1.781]

S6 Interpretation and access

All five quantitative transformations are retained; combined applies removal, renaming and formatting, without injection. Parsing checks do not prove semantic equivalence. Per-language transformed scores, change fractions and parser-regression statistics remain in the accompanying attack decomposition CSV. Full grouped importance and ranking stability are available in the local mechanism CSVs. These materials are included in the publicly accessible research deposit.

Code, pinned dependencies, frozen records, figure source data and reproduction commands are publicly deposited at <https://github.com/cdhuangjin/lpcode-mstf-leakage-safe-detection/tree/1b6a5b9f7f274b22a718b53219581d5f57a30792>. Anonymous access was verified on 5 September 2026. Raw LPcode data are obtained from its original pinned repository, with checksums supplied in the deposit. Open-reuse licensing remains subject to author confirmation; no archival DOI or independent formal retraining is claimed.